

## Math 421

Wednesday, December 3

## Announcements

- Drop-In Hours
  - **Today** 12-1 pm VV 311
  - Thursday 9-11 am VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
- Homework 10 due Friday (last HW!)

# Another Corollary of the MVT

**Corollary 29.7.** Let  $f$  be a differentiable function on an interval  $(a, b)$ .

1. If  $f'(x) > 0$  for all  $x \in (a, b)$ , then  $f$  is strictly increasing.
2. If  $f'(x) < 0$  for all  $x \in (a, b)$ , then  $f$  is strictly decreasing.

Proof of 1:

# Motivating Question

**Q:** Is the derivative of a differentiable function always continuous?

# Example

Let

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then

$$f'(x) = \begin{cases} -\cos \frac{1}{x} + 2x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

# Intermediate Value Property for Derivatives

**Theorem 29.8.** Let  $f$  be a differentiable function on  $(a, b)$ . Then  $f'(x)$  has the intermediate value property on  $(a, b)$ . That is:

If  $a < x_1 < x_2 < b$ , and if  $c$  lies between  $f'(x_1)$  and  $f'(x_2)$ , there exists (at least one)  $x_0$  in  $(x_1, x_2)$  such that  $f'(x_0) = c$ .

# Chapter 6: Integration

Section 32: The Riemann Integral

# Integration – Informal Idea

# Partitions

**Def.** Let  $a < b$ . A **partition** of the interval  $[a, b]$  is a finite collection of points in  $[a, b]$ ,  $P = \{t_0, t_1, \dots, t_{n-1}, t_n\}$  such that

$$a = t_0 < t_1 < \dots < t_{n-1} < t_n = b.$$

# Upper & Lower Sums

**Def.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be bounded and  $P = \{t_0, t_1, \dots, t_{n-1}, t_n\}$  be a partition of  $[a, b]$ . The **upper and lower (Darboux) sums** for  $P$  are defined by

$$U(f, P) = \sum_{k=1}^n M_k (t_k - t_{k-1}) \quad M_k = \sup\{f(x) : x \in [t_{k-1}, t_k]\}$$

$$L(f, P) = \sum_{k=1}^n m_k (t_k - t_{k-1}) \quad m_k = \inf\{f(x) : x \in [t_{k-1}, t_k]\}$$

# Divide & Conquer

Let  $f(x) = x^2$ . On the interval  $[-1, 4]$  find the upper and lower Darboux sums for the given partitions:

1.  $Q = \{-1, 0, 2, 4\}$

a) Upper sum

b) Lower sum

2.  $P = \{-1, 4\}$

a) Upper sum

b) Lower sum

# Activity

Let  $c \in \mathbb{R}$  be a constant. Find the upper and lower sums for the following functions on  $[-1,1]$  for an arbitrary partition  $P$ .

1. Define  $f(x) = c$ .

2. Define  $g(x) = \begin{cases} c, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$

# Discussion Questions

- 1 Why must  $f$  be bounded?
- 2 Why do we define  $m_k$  and  $M_k$  in terms of  $\inf$  and  $\sup$  ?
- 3 Is  $L(f, P) \leq U(f, P)$ ? Make a guess as to whether  $L(f, P_1) \leq U(f, P_2)$  for any two partitions  $P_1$  and  $P_2$ .

## 32.2 Lemma

**Lemma.** If  $P, Q$  are partitions of  $[a, b]$  such that  $P \subseteq Q$  and  $f$  is bounded on  $[a, b]$ , then

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$$

**Activity:** Convince yourself this theorem is true, perhaps by drawing a picture.

## 32.3 Lemma

**Lemma.** Let  $P_1$  and  $P_2$  be partitions of  $[a, b]$  and let  $f$  be a function which is bounded on  $[a, b]$ . Then  $L(f, P_1) \leq U(f, P_2)$ .

**Proof.**

1. Define an appropriate new set  $Q$ .
2. Apply Lemma 32.2 to  $P_1, P_2$  and  $Q$ .

# Corollary

Def.

- $U(f) = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$
- $L(f) = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}$

**Cor.** If  $f$  is bounded on  $[a, b]$ , then  $L(f) \leq U(f)$ .

# Integrable

**Def.** A function  $f$  which is bounded on  $[a, b]$  is **integrable** on  $[a, b]$  if

$$L(f) = U(f)$$

In this case, the value is called the **integral** of  $f$  and is denoted by

$$\int_a^b f = \int_a^b f(x) dx.$$

# Activity

Determine whether or not the following functions are integrable on  $[-1,1]$ . Let  $c \in \mathbb{R}$  be a constant.

1. Define  $f(x) = c$ .

2. Define  $g(x) = \begin{cases} c, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$

# Equivalent Definition

**Thm.** (“Cauchy criterion for integrability”) If  $f$  is bounded on  $[a, b]$ , then  $f$  is integrable on  $[a, b]$  if and only if for every  $\epsilon > 0$ , there exists a partition  $P$  of  $[a, b]$  such that

$$U(f, P) - L(f, P) < \epsilon$$

**Activity:** Convince yourself this is an equivalent definition.

# Activity

Determine whether or not the following function is integrable on  $[-1,1]$

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$